Statistics for Data Analysts

Unit 14: A/B testing

Google Reichman University | Google and Reichman Tech School

Units overview

Description

This lesson focuses on the principles and methodology of A/B testing, a statistical technique commonly used in data analysis to compare two or more variants and determine their impact on user behavior or outcomes.

Goals and outcomes

The goal of this lesson is to familiarize you with the concept of A/B testing and provide a comprehensive understanding of its purpose, process, and key considerations. By the end of this lesson, you will be able to design and implement A/B tests effectively, analyze the results using statistical methods, and make data-driven recommendations based on their findings.

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Introduction to A/B testing

2:00

| Key Principles of A/B Testing

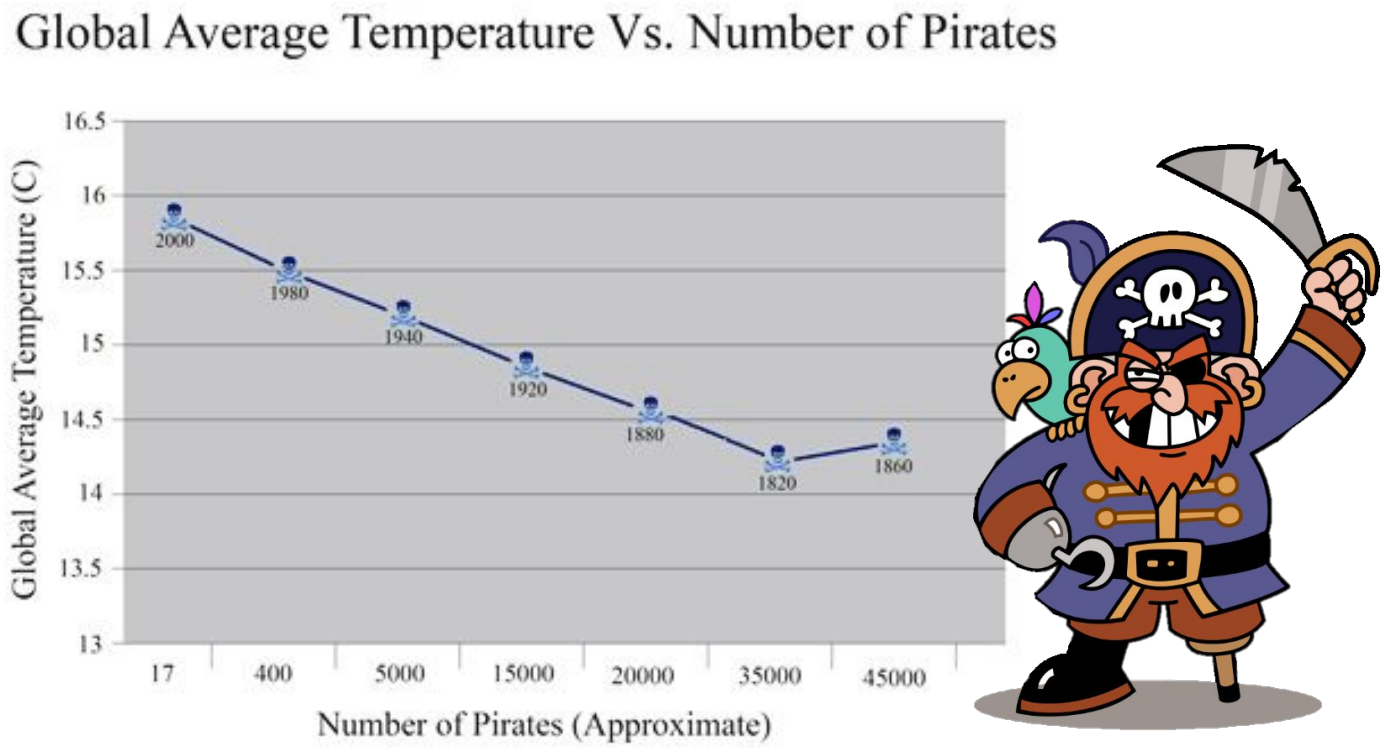
| Designing an A/B Test

| A/B Testing: an example

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Introduction to A/B testing

- WHAT CAUSES WHAT?



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Introduction to A/B testing

- We want to understand causation to make an informed decision.
- Separating causation from association can be a serious challenge
- For example: **does changing the color of a button on a website lead to a higher click-through rate?**



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Reminder…

Introduction to A/B testing

We talked about two types of research:

- Observational research
- Experimental research

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Reminder…

Introduction to A/B testing

*From Unit 01

Observational research:

- Involves observing and measuring a phenomenon or behavior without manipulating any variables.
- The researcher does not intervene in the natural setting or behavior, and collects data through observation, surveys, or other methods.
- The goal is to describe or understand the behavior or phenomenon being studied.
- *Disadvantage:* great difficulty in assuming cause-and-effect

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Reminder…

Introduction to A/B testing

*From Unit 01

Experimental research:

- Involves manipulating one or more variables to observe the effect on the dependent variable.
- Participants are randomly assigned to different groups. The researcher manipulates the independent variable in one or more of the groups and measures the effects on the dependent variable.
- The goal is to establish cause-and-effect relationships.

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Introduction to A/B testing

- **A/B testing** is the practice of using of randomized experiments for making informed decisions that rely on understanding cause and effect
- It plays a vital role in evaluating the effectiveness of different strategies, designs, or interventions.
- A/B testing systematically compares multiple variants and analyzes user responses, therefore helps us make data-driven decisions and optimize outcomes.



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Key Principles in designing A/B Testing

1. Defining our goal:

Before conducting A/B testing, we define what our goal is and what we try to achieve.

- Higher engagement on the app
- Spending more money on shopping site
- Clicking more often / faster
- Etc…

The end-goal is to understand whether this change is beneficial for our purposes… or not!



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Key Principles in designing A/B Testing

Introduction to A/B testing

2. Formulate clear Hypotheses:

A/B testing requires formulating well-defined hypotheses about the expected impact of different variants on user behavior or outcomes.

- **H_0:** There is no significant difference in the click-through rates between the blue button **(Variant A)** and the green button **(Variant B)**.
- **H_1:** Changing the button color from blue **(Variant A)** to green **(Variant B)** will lead to a higher click-through rate.



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Key Principles in designing A/B Testing

3. Isolate variables:

To make sure our variation is the only cause for difference, we have to avoid any other change.

For example, if we also want to consider the influence of the size of the button, we will not change both size & color at the same time.

For each variable, we will conduct an experiment of its own



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Key Principles in designing A/B Testing

4. Planning ahead:

We will decide in advance all the important details:

- Research question
- How are we going to manipulate the variable
- What are the criterions to reject H_0
- What size the sample will be
- How long the experiment is going to take



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Key Principles in designing A/B Testing

Introduction to A/B testing

5. Control Group Selection:

We will include a control group to establish a baseline and accurately measure the impact of the treatment.

6. Randomization:

Randomly assign participants to different variants to ensure unbiased and fair testing conditions.



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KPIs

Introduction to A/B testing

- As A/B testing helps us make informed decisions about product or process changes, **Key Performance Indicators (KPIs)** are crucial metrics that measure the impact of these changes:
  - KPIs are quantifiable, and thus, they can measure the success or performance of a particular goal.
  - Ultimately, KPIs are what help us determine which variant performs better.

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Common KPIs

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1. Conversion Rate:

   Measures the percentage of users who complete a desired action (e.g., purchase, sign-up) out of the total visitors.

   Calculated as conversions divided by total visitors.

2. Click-Through Rate (CTR):

   Measures the percentage of users who click on a specific element (e.g., button, link) out of the total users who view it.

   Calculated as clicks divided by impressions.

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Common KPIs

Introduction to A/B testing

3. Revenue per User (RPU):
   Measures the average revenue generated per user.
   Calculated as total revenue divided by the number of users.

4. Average Order Value (AOV):
   Measures the average value of each order placed by users.
   Calculated as total revenue divided by the number of orders.

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Common KPIs

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5. Bounce Rate:

   Measures the percentage of users who visit a page and leave without taking any further action.

   Calculated as the number of bounces divided by the number of visits.

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Analyzing KPIs

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Statistical Significance:

- We will use statistical analysis to determine whether observed differences between variants are **statistically** meaningful.
- The appropriate statistical tool will be determined according to the research question, the manipulation, and the variable that is being used



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Analyzing KPIs

Introduction to A/B testing

Statistical tools:

For example:

- If we measure the **reaction time** for different colors among independent groups, we will use **t-tests for independent samples**
- If we measure the **gender** of users (whether it is different than 50/50), we will use **Chi-squared goodness of fit.**

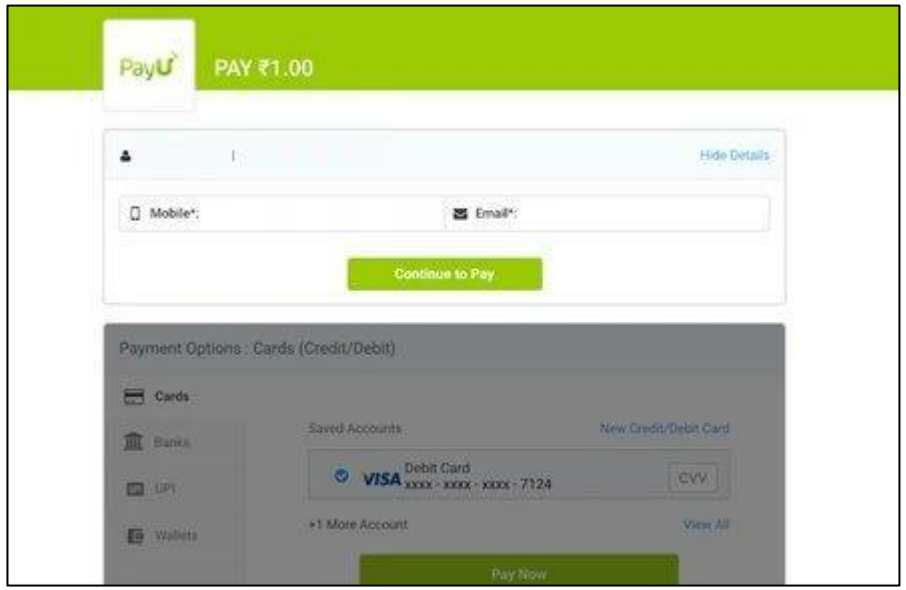


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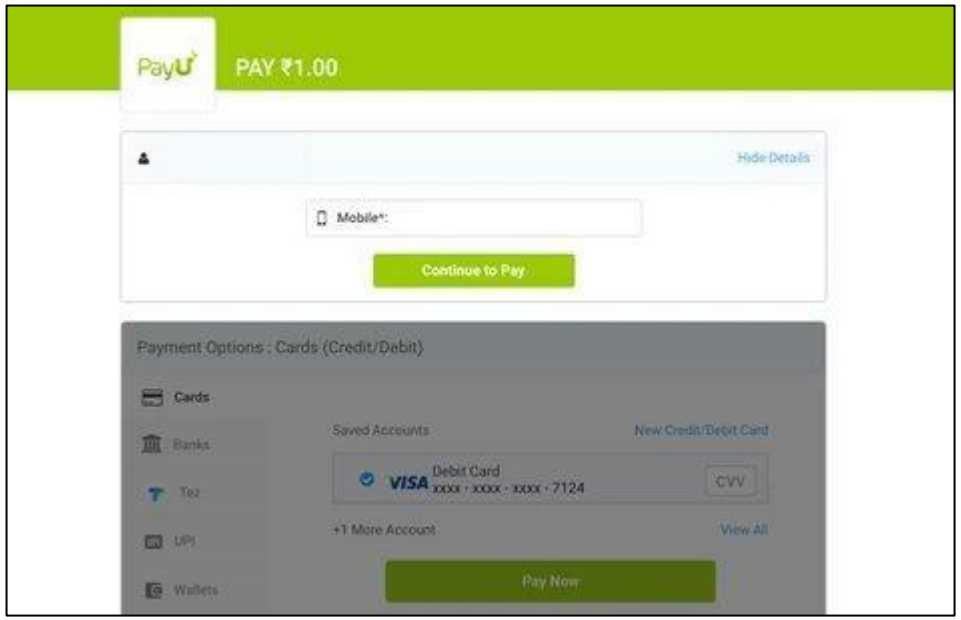
A/B testing: an example

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Control



Variation



Increased conversions by 5.8%

https://vwo.com/blog/ab-testing-examples/

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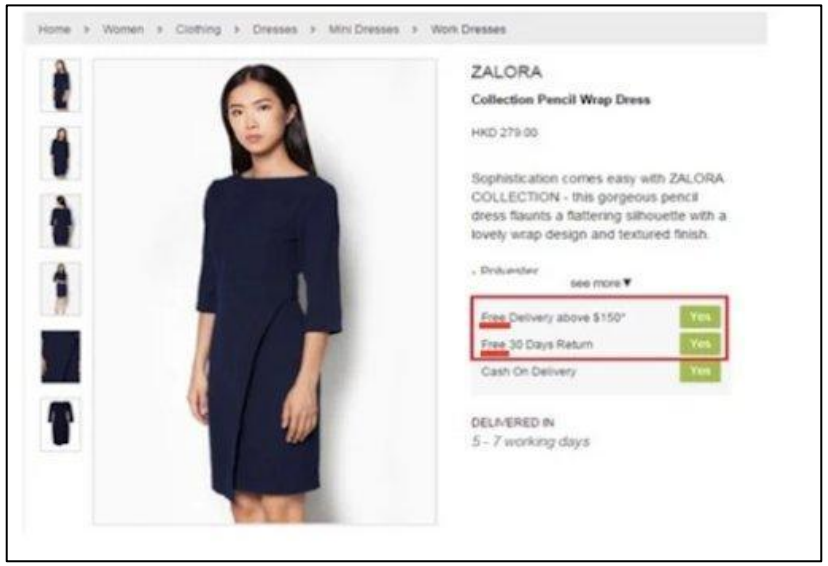
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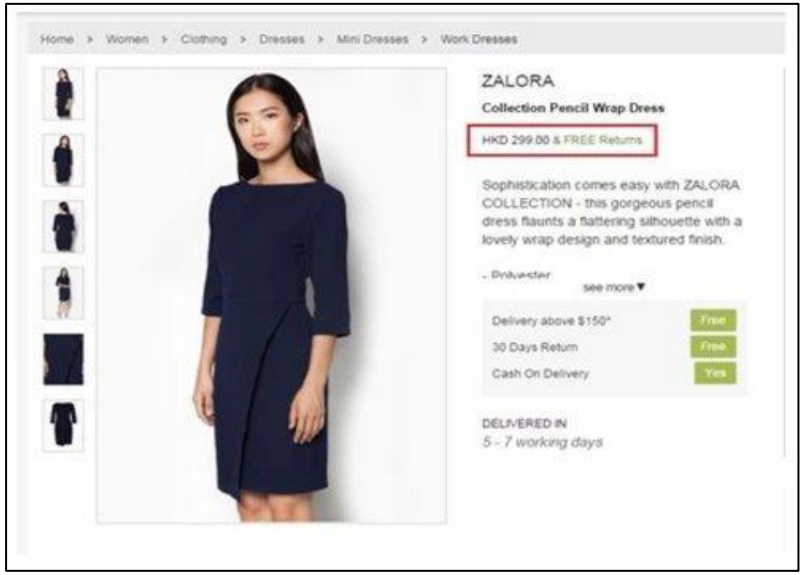
Control



Variation 1



Variation 2



Increased checkout rate by 12.3%

https://vwo.com/blog/ab-testing-examples/

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Let’s play

Introduction to A/B testing

Let’s open a file and simulate an A/B test…

https://docs.google.com/spreadsheets/d/1QJL2LR86Vejh8sOmC6jyv0MxiwGrvsrAeekYI47VXgE/edit?usp=sharing

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